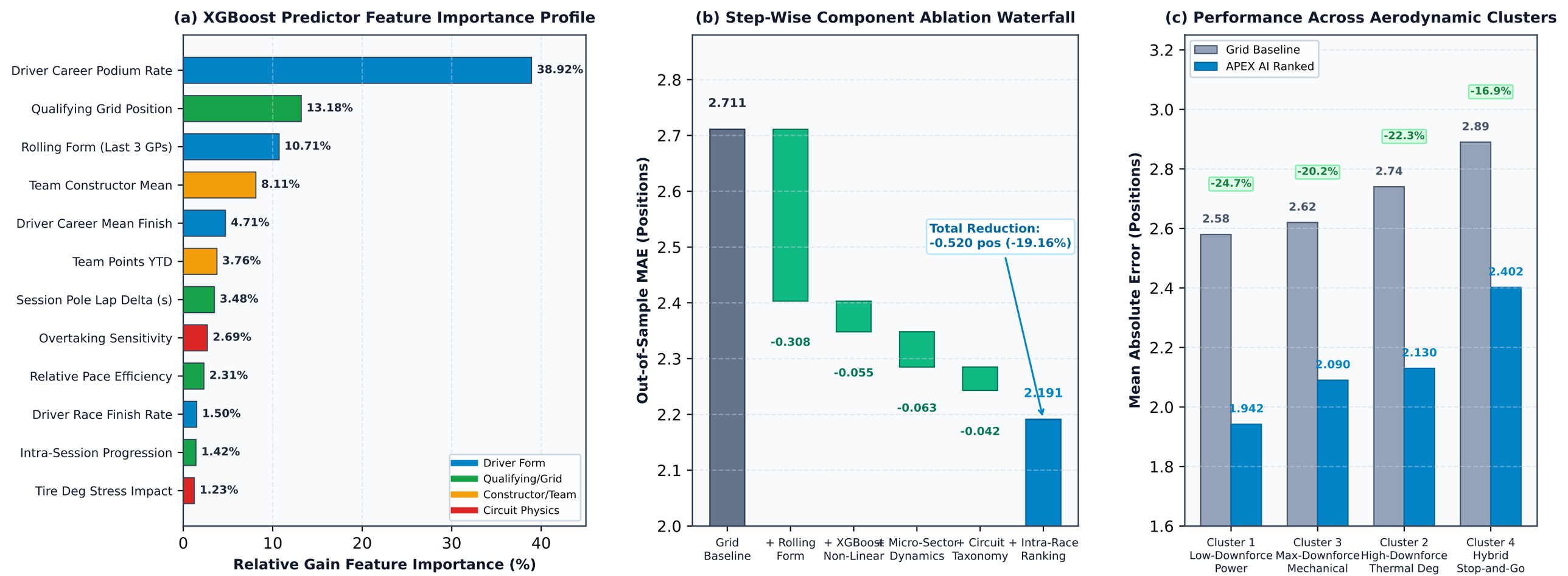
(a) XGBoost Predictor Feature Importance Profile
Driver Career Podium Rate
Qualifying Grid Position
Rolling Form (Last 3 GPs)
Team Constructor Mean
Driver Career Mean Finish
Team Points YTD
Session Pole Lap Delta (s)
Overtaking Sensitivity
Relative Pace Efficiency
Driver Race Finish Rate
Intra-Session Progression
Tire Deg Stress Impact
38.92%
13.18%
10.71%
8.11%
4.71%
3.76%
3.48%
2.69%
2.31%
1.50%
1.42%
1.23%
Driver Form
Qualifying/Grid
Constructor/Team
Circuit Physics
0
10
20
30
40
Relative Gain Feature Importance (%)
(b) Step-Wise Component Ablation Waterfall
Out-of-Sample MAE (Positions)
2.0
2.1
2.2
2.3
2.4
2.5
2.6
2.7
2.8
2.711
-0.308
-0.055
-0.063
-0.042
2.191
Total Reduction:
-0.520 pos (-19.16%)
Grid Baseline
+ Rolling Form
+ XGBoost Non-Linear
+ Micro-Sector Dynamics
+ Circuit Taxonomy
+ Intra-Race Ranking
(c) Performance Across Aerodynamic Clusters
Mean Absolute Error (Positions)
1.6
1.8
2.0
2.2
2.4
2.6
2.8
3.0
3.2
Grid Baseline
APEX AI Ranked
-24.7%
-20.2%
-22.3%
-16.9%
2.58
1.942
2.62
2.090
2.74
2.130
2.89
2.402
Cluster 1
Low-Downforce
Power
Cluster 3
Max-Downforce
Mechanical
Cluster 2
High-Downforce
Thermal Deg
Cluster 4
Hybrid
Stop-and-Go